

The free group algebra $\ell^1(\mathbb{F}_2)$ is not ideally amenable

Candidate full counterexample packet for arXiv:math/0503093

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Abstract

Let $q : \mathbb{F}_2 \rightarrow \mathbb{Z}^2$ be abelianization and let I be the kernel of the induced quotient homomorphism $\ell^1(\mathbb{F}_2) \rightarrow \ell^1(\mathbb{Z}^2)$. We construct a non-inner bounded derivation from $\ell^1(\mathbb{F}_2)$ into I^* . The obstruction is a scalar bounded two-cocycle coming from an explicit split quasimorphism. Placing that cocycle in a nonidentity abelianization fiber makes it nonzero in $I^\perp$, but zero after inclusion in the full conjugation module $\ell^\infty(\mathbb{F}_2)$. The resulting connecting class is the required derivation. This gives a negative answer to Question 4.4 of the source.

Status. This is a candidate full proof. The bounded-cohomology inputs used below are standard and stated explicitly; the application to the abelianization ideal is the proposed new step. No claim is made that the surrounding general classification problem for $L^1(G)$ is solved.

1 The ideal and the coefficient modules

Put $G = \mathbb{F}_2 = \langle a \rangle * \langle b \rangle$, $H = G_{\text{ab}} \cong \mathbb{Z}^2$, and let $q : G \rightarrow H$ be the quotient map. The contractive surjective algebra homomorphism

$$Q : \ell^1(G) \longrightarrow \ell^1(H), \quad (Qf)(h) = \sum_{q(g)=h} f(g),$$

has a closed two-sided kernel I . Write

$$E = \ell^\infty(G) = \ell^1(G)^*, \quad Y = I^\perp = Q^* \ell^\infty(H), \quad X = E/Y \cong I^*.$$

Thus Y consists exactly of the functions $z \circ q$. Give these spaces the conjugation action

$$(g \star F)(x) = F(g^{-1}xg). \tag{1}$$

Since H is abelian, the action on $Y \cong \ell^\infty(H)$ is trivial. The sequence

$$0 \longrightarrow Y \xrightarrow{j} E \xrightarrow{p} X \longrightarrow 0 \tag{2}$$

is admissible as a sequence of Banach spaces: a set-theoretic section $s : H \rightarrow G$ gives the norm-one retraction $F \mapsto (h \mapsto F(s(h)))$ of j .

We use inhomogeneous bounded cochains, with

$$(\partial\xi)(g) = g \star \xi - \xi, \quad (\partial U)(g, h) = g \star U(h) - U(gh) + U(g).$$

Lemma 1 (derivations and conjugation cocycles). *For every closed ideal J of $\ell^1(G)$, bounded derivations $D : \ell^1(G) \rightarrow J^*$ correspond to bounded one-cocycles $b : G \rightarrow J^*$ for (1), via*

$$b(g) = D(\delta_g) \cdot \delta_{g^{-1}}, \quad D(\delta_g) = b(g) \cdot \delta_g.$$

Inner derivations correspond exactly to coboundaries.

Proof. The Leibniz rule gives

$$b(gh) = b(g) + g \star b(h).$$

Conversely this identity gives $D(\delta_g \delta_h) = \delta_g \cdot D(\delta_h) + D(\delta_g) \cdot \delta_h$. Since the module actions of point masses are isometric and b is bounded, the formula extends boundedly from point masses to $\ell^1(G)$. Finally, $D(\delta_g) = \delta_g \cdot \xi - \xi \cdot \delta_g$ is equivalent to $b(g) = g \star \xi - \xi$. $\square$

2 The cohomological mechanism

We isolate the only standard bounded-cohomology input.

Lemma 2 (amenable-stabilizer vanishing). *If a discrete group Γ acts on a set Ω and every point stabilizer is amenable, then*

$$H_b^n(\Gamma, \ell^\infty(\Omega)) = 0 \quad (n \geq 1).$$

The assertion remains valid for a disjoint union of arbitrarily many orbits.

Proof. For a transitive orbit Γ/K , bounded Shapiro identifies the displayed group with $H_b^n(K, \mathbb{C})$. Amenability of K supplies a contracting homotopy of norm one, hence the latter group vanishes. The Shapiro maps and the amenability contraction are norm-controlled independently of the orbit. For a disjoint union, apply them orbitwise and take the resulting ℓ^∞ -product; the uniform norm bound makes the combined primitive bounded. This is the standard coinduced-module proof of bounded Shapiro and amenable vanishing [2, Chs. 7, 10]. $\square$

Let

$$E_0 = \{F \in \ell^\infty(G) : F(e) = 0\} = \ell^\infty(G \setminus \{e\}).$$

Every nontrivial element of a free group has infinite cyclic centralizer. The conjugation action on $G \setminus \{e\}$ therefore has amenable stabilizers. Lemma 2 yields

$$H_b^n(G, E_0) = 0 \quad (n \geq 1). \quad (3)$$

We now construct the obstruction. Write each nonidentity element of $G = \langle a \rangle * \langle b \rangle$ in reduced syllable form and define

$$\varphi(g) = \sum_{\substack{\text{syllables } a^k \\ \text{in the reduced form of } g}} \text{sgn}(k), \quad \varphi(e) = 0.$$

When two reduced words are multiplied, fully cancelled syllable pairs make opposite contributions to φ , and at most one final pair of syllables is merged. Consequently

$$c(g, h) := (\partial\varphi)(g, h) = \varphi(g) + \varphi(h) - \varphi(gh)$$

is bounded (indeed $|c(g, h)| \leq 3$). Its class in $H_b^2(G, \mathbb{C})$ is nonzero. For if $c = \partial u$ with u bounded, then $\varphi - u$ would be a homomorphism. But

$$\frac{\varphi(a^n)}{n} \rightarrow 0, \quad \frac{\varphi(b^n)}{n} = 0, \quad \frac{\varphi((ab)^n)}{n} = 1,$$

which is impossible for a homomorphism at bounded distance from φ .

Let $h_0 = (1, 0) \in H$, and put $z = \mathbf{1}_{\{h_0\}} \in \ell^\infty(H)$. Because Y is a trivial G -module, cz is a bounded Y -valued two-cocycle. It is not a coboundary: evaluation at h_0 is an equivariant bounded functional and sends cz to the nonzero class c .

On the other hand, the E -valued cocycle $j(cz)$ is supported on the fiber $q^{-1}(h_0)$, which does not contain e . It therefore takes values in E_0 , and (3) supplies a bounded one-cochain $U : G \rightarrow E_0$ such that

$$\partial U = j(cz). \quad (4)$$

Set $b = pU : G \rightarrow X$. Equation (4) gives $\partial b = 0$. Moreover, b is not a coboundary. Indeed, if $b = \partial\xi$, choose a lift $\tilde{\xi} \in E$ of ξ . Then $U - \partial\tilde{\xi}$ is a bounded Y -valued one-cochain, and its coboundary is $j(cz)$. This would make cz a coboundary in Y , a contradiction. Hence

$$[b] \neq 0 \quad \text{in} \quad H_b^1(G, X). \quad (5)$$

3 The counterexample

Theorem 3. *Let I be the kernel of the abelianization quotient*

$$\ell^1(\mathbb{F}_2) \longrightarrow \ell^1(\mathbb{Z}^2).$$

Then

$$H^1(\ell^1(\mathbb{F}_2), I^*) \neq 0.$$

In particular, $\ell^1(\mathbb{F}_2)$ is not ideally amenable.

Proof. Identify I^* with X , and let b be the nonzero bounded cocycle from (5). Define first on point masses

$$D(\delta_g) = b(g) \cdot \delta_g.$$

Lemma 1 extends this to a bounded derivation $D : \ell^1(G) \rightarrow I^*$. If D were inner, the same lemma would make b a coboundary, contradicting (5). $\square$

Remark 4 (a wider criterion). *The same proof works for any discrete group G which has a nontrivial abelian quotient, amenable centralizers of all nonidentity elements, and $H_b^2(G, \mathbb{C}) \neq 0$. The free group supplies the cleanest instance.*

4 Source evidence

The source's page 407 asks explicitly whether $\ell^1(F_2)$ is ideally amenable:

Question 4.1. If $\mathcal{A}$ and $\mathcal{B}$ are ideally amenable Banach algebras, then is $\mathcal{A} \widehat{\otimes} \mathcal{B}$ ideally amenable?

We know that $L^1(G)$ is amenable if and only if G is an amenable group [Jo1], and also $L^1(G)$ is weakly amenable for every locally compact group ([Jo1] or [D-Gh]).

Question 4.2. Under what conditions the group algebra $L^1(G)$ is ideally amenable?

Question 4.3. If $L^1(G)^{**}$ is ideally amenable, then so are $M(G)$ and $L^1(G)$ ideally amenable?

Question 4.4. Is $\ell^1(F_2)$ ideally amenable?

For a Banach algebra $\mathcal{A}$, the amenability of $\mathcal{A}^{**}$ necessitates the amenability of $\mathcal{A}$ ([Da], Proposition 2.8.59) and similarly for weak amenability provided $\mathcal{A}$ is a left ideal in $\mathcal{A}^{**}$ [Gh-L-W]. So we can raise the following question.

Question 4.5. If $\mathcal{A}^{**}$ is ideally amenable, then is $\mathcal{A}$ ideally amenable?

References

- [1] M. E. Gordji and T. Yazdanpanah, *Derivations into duals of ideals of Banach algebras*, Proc. Indian Acad. Sci. Math. Sci. **114** (2004), 399–408; arXiv:math/0503093. Question 4.4 is on p. 407.
- [2] N. Monod, *Continuous Bounded Cohomology of Locally Compact Groups*, Lecture Notes in Mathematics 1758, Springer, 2001. See the bounded Shapiro lemma, relatively injective/coinduced modules, and amenable vanishing.
- [3] N. Grønbaek and A. T.-M. Lau, *On Hochschild cohomology of the augmentation ideal of a locally compact group*, Math. Proc. Cambridge Philos. Soc. **126** (1999), 139–148. This treats the ordinary augmentation ideal and explains why that ideal alone does not settle the source question.